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CIS5500 Group 13
Final Project Milestone 3

Host: database-1.cv1pozpjpynb.us-east-1.rds.amazonaws.com
Port: 5432
Username: cis5500group13
Password: kasbik-cegje4-tyFsyf
Db: postgres

We will organize the data into three connected tables:
region: stores the canonical geographic entity (region_id, region name, state, region type).
acs_income: stores annual ACS income information by region (region_id, year, household_income, num_households).
zhvi_value: stores Zillow home value time series by region (region_id, date, value).

Queries 1-10:

Query 1: Count Regions
Description: Counts the total number of regions in the dataset.

```
SELECT COUNT(*) AS total_regions
FROM region;
```

Query 2: Average Income by Year
Description: Computes the average median household income across all regions for each year.

```
SELECT year, AVG(household_income) AS avg_income
FROM acs_income
GROUP BY year
ORDER BY year;
```

Query 3: Latest Home Values
Description: Retrieves the most recent home value available for each region.

```
SELECT region_id, value
FROM zhvi_value
WHERE date = (SELECT MAX(date) FROM zhvi_value);
```

Query 4: Top 25 Most Expensive Regions
Description: Finds the 25 regions with the highest home values in the most recent period.

```
SELECT r.region_id, r.region_name, r.state_name, z.value
FROM zhvi_value z
JOIN region r ON z.region_id = r.region_id
WHERE z.date = (SELECT MAX(date) FROM zhvi_value)
ORDER BY z.value DESC
LIMIT 25;
```

Query 5: Income Ranking
Description: Ranks regions by median household income in the most recent year available.

```
SELECT r.region_id,
r.region_name,
r.state_name,
a.household_income,
RANK() OVER (ORDER BY a.household_income DESC) AS income_rank
FROM acs_income a
JOIN region r ON a.region_id = r.region_id
WHERE a.year = (SELECT MAX(year) FROM acs_income);
```

Query 6: Average Home Value by State
Description: Computes the average home value for each state using the most recent available housing data.

```
SELECT r.state_name, AVG(z.value) AS avg_home_value
FROM zhvi_value z
JOIN region r ON z.region_id = r.region_id
WHERE z.date = (SELECT MAX(date) FROM zhvi_value)
GROUP BY r.state_name
ORDER BY avg_home_value DESC;
```

Complex Queries

Query 7: Affordability Ratio Ranking
Description: Calculates the price-to-income ratio for each region using the latest income and housing data, then ranks regions by affordability.

```
WITH latest_income AS (
SELECT region_id, household_income
FROM acs_income
WHERE year = (SELECT MAX(year) FROM acs_income)
),
latest_home AS (
SELECT region_id, value
FROM zhvi_value
```

```
WHERE date = (SELECT MAX(date) FROM zhvi_value)
)
SELECT r.region_id,
r.region_name,
r.state_name,
l.value,
i.household_income,
ROUND((l.value / NULLIF(i.household_income, 0))::numeric, 2) AS affordability_ratio
FROM latest_income i
JOIN latest_home l ON i.region_id = l.region_id
JOIN region r ON r.region_id = i.region_id
ORDER BY affordability_ratio DESC
LIMIT 25;
```

Query 8: High-Income Regions with Poor Affordability
Description: Identifies regions in the top income quartile that still have high home-price-to-income ratios.

```
WITH latest_income AS (
SELECT region_id,
household_income,
NTILE(4) OVER (ORDER BY household_income DESC) AS income_quartile
FROM acs_income
WHERE year = (SELECT MAX(year) FROM acs_income)
),
latest_home AS (
SELECT region_id, value
FROM zhvi_value
WHERE date = (SELECT MAX(date) FROM zhvi_value)
)
SELECT r.region_id,
r.region_name,
r.state_name,
i.household_income,
h.value,
ROUND((h.value / NULLIF(i.household_income, 0))::numeric, 2) AS affordability_ratio
FROM latest_income i
JOIN latest_home h ON i.region_id = h.region_id
JOIN region r ON r.region_id = i.region_id
WHERE i.income_quartile = 1
ORDER BY affordability_ratio DESC;
```

Query 9: Previous-Period Home Value Growth

Description: Computes the percentage change in home values from the previous recorded period for each region.

```
SELECT r.region_id,
r.region_name,
r.state_name,
z.date,
z.value,
LAG(z.value) OVER (PARTITION BY z.region_id ORDER BY z.date) AS prev_value,
ROUND(
(
(z.value - LAG(z.value) OVER (PARTITION BY z.region_id ORDER BY z.date))
/ NULLIF(LAG(z.value) OVER (PARTITION BY z.region_id ORDER BY z.date), 0)
)::numeric * 100,
2
) AS pct_growth
FROM zhvi_value z
JOIN region r ON z.region_id = r.region_id
ORDER BY z.region_id, z.date;
```

Query 10: Similar Regions Based on Income and Home Value
Description: Finds regions with household income and home values most similar to a selected target region.

```
WITH target AS (
SELECT i.region_id,
i.household_income,
z.value
FROM acs_income i
JOIN zhvi_value z ON i.region_id = z.region_id
WHERE i.year = (SELECT MAX(year) FROM acs_income)
AND z.date = (SELECT MAX(date) FROM zhvi_value)
AND i.region_id = 12345
),
all_regions AS (
SELECT i.region_id,
r.region_name,
r.state_name,
i.household_income,
z.value
FROM acs_income i
JOIN zhvi_value z ON i.region_id = z.region_id
JOIN region r ON i.region_id = r.region_id
WHERE i.year = (SELECT MAX(year) FROM acs_income)
```

```
AND z.date = (SELECT MAX(date) FROM zhvi_value)
)
SELECT a.region_id,
a.region_name,
a.state_name,
a.household_income,
a.value,
ABS(a.household_income - t.household_income) +
ABS(a.value - t.value) AS similarity_score
FROM all_regions a
CROSS JOIN target t
WHERE a.region_id <> t.region_id
ORDER BY similarity_score ASC
LIMIT 10;
```